Harish G

harishg.hari001@gmail.com | 9361060129
github.com/harishg2001 | linkedin.com/in/harish-g-6a03402b6

Professional Summary

Embedded Software Engineer with 1 year of Experience specializing in firmware development and real-time systems. Proven ability to deliver reliable embedded solutions and a strong commitment to continuous learning and professional growth.

Skills

Languages: C/C++, RUST, Embedded C, Linux System Programming, RTOS

Operating System: Linux

Communication Protocols: UART, I2C, SPI, CAN

Networking Protocols: HTTP, HTTPS, REDFISH, SSH/CLI, FTPS, SNMP, SMTP

Development Tools: JIRA, GitKraken, Eclipse, GCC, GDB, Keil IDE

Project Tools:POSTMAN, PUTTY, MIB BROWSER

Hardware: Network Management Controller, PDU, LPC2129, ARM7

Work Experience

nVent, Bangalore Feb 2025 - Present

Embedded Software Engineer

A company specialized in innovative electrical connection solution

- Engaged with customers to assess and address critical hot fix and security issues impacting their sites.
- Managed future firmware release planning, ensuring timely deployment to production environment or direct delivery of targeted hot-fixes to affected customer sites.
- Evaluated and prioritized tasks and based on severity, considering feedback from daily debugging sessions.
- Performed meticulous code reviews, reducing bugs and fostering a culture of continuous improvement.
- Utilized Agile methodology and jira for task management and sprint planning, which boosted productivity and ensured timely project delivery.

Vector India, Bangalore Nov 2024 - Feb 2025

Embedded Lab Coordinator

- Guided newly enrolled trainees in C, C++, Embedded C, and Linux IPC concepts during scheduled lab sessions.
- Provided hands-on support with debugging, syntax correction, and logic building for embedded programming exercises.

Vector India, Bangalore Feb 2024 - Sep 2024

Embedded Systems Trainee

- Completed intensive hands-on training in Embedded C, Data Structures, RTOS, and Micro controller Programming.
- Worked on real-time projects involving UART, SPI, I2C, and CAN protocols.
- Gained practical experience in firmware debugging, interrupt handling, and memory management.
- Developed mini-projects using Keil simulating real-world embedded applications.
- Learned to interface peripherals like LCDs, ADC/DAC, and sensors with micro controllers.
- Implemented Linux IPC mechanisms including Pipe, FIFO, Message queue, Shared memory, Semaphore, Mutex and threads.

Education

Government College of Engineering, Erode Aug 2019 - Jun 2023

B.E. in Electronics and Communication Engineering ***CGPA: 7.5/10***

Focused on core electronics and programming fundamentals with exposure to embedded systems and software development.

Project Work

Enlogic Power Distribution Unit Organization: nVent

- Worked as part of the Firmware Development team at nVent, involved in the development and maintenance of firmware for intelligent Unit. These devices are equipped with multiple management interfaces including WEBUI, SNMP, Redfish,, CLI, SSH, and TELNET. The PDUs are designed for the use in data centers, offering advanced monitoring and remote management capabilities.
- Experinence with libraries: CJSON, Jannson, Curl, SSL, Zlib, WPA-Client.Configured nVent Products like Enlogic, Rackpower, Dellipv6 PDUs.Sensor integration experience Temperature, Humidity, Door, Smoke,dry, Spot fluid.
- Performed issue triage, root cause analysis, bug fixing , feature implementation ensure system reliability and performance
- Conducted developer-level testing, regression verification, and code reviews for continuous improvement.
- Worked with RESTful APIs and system level APIs to enhance device integration and remote management.

Employee Attendance System Using RFID and LINUX Organization: Vector India

- Developed the attendance system using LPC2129 Micro controller, RFID, UART, I2C.
- This system leverage Radio frequency Identification(RFID) technology to record attendance data and employs a Linux based server for data management, storage, and processing.
- I2C -base Real Time Clock(RTC) module for accurate timekeeping and UART protocol for transmit the data from Micro controller to Monitor using Linux

Student Database Management System (Data structure using C Programming) Organization: Vector India

- Designed a system to manage student records using single linked list for efficient data handling
- Maintain the student data record accordingly to roll number,name, mark and used file system to store the data
- Features included adding, viewing, searching, modifying, sorting and deleting records.

Smart Home Automation Using UART Organization: Vector India

- Controlled home Appliances via a mobile connected through an HC05 Bluetooth module.
- Successfully interfaced the module the LPC2129 Micro controller.

Training and Certificates

- Completed Advanced Embedded System Course at vector India Pvt.Ltd, Bengalure.
- Lab Coordinator at vector India Pvt.Ltd, Bengalure.